

Multiple Choice (26 marks)

1. Which of the following facts was established prior to 1859, the year in which Charles Darwin published *On the Origin of Species*?
 - (a) DNA provides the macromolecular basis of heredity.
 - (b) Mendelian principles explain why some traits are dominant and others are recessive.
 - (c) Prokaryotes include two major domains, the Bacteria and the Archaea.
 - (d) There exist fossilized remains of species that have become extinct.
2. Which of the following would increase the rate at which a gas diffuses between the alveoli of the lungs and the pulmonary capillaries?
 - (a) Decreasing the partial pressure gradient of the gas
 - (b) Decreasing the solubility of the gas in water
 - (c) Increasing the total surface area available for diffusion
 - (d) Decreasing the rate of blood flow through the pulmonary capillary
3. Short-term changes in plant growth rate mediated by the plant hormone auxin are hypothesized to involve:
 - (a) loss of turgor pressure in the affected cells
 - (b) increased extensibility of the walls of affected cells
 - (c) suppression of metabolic activity in affected cells
 - (d) cytoskeletal rearrangements in the affected cells
4. Targeting of a newly synthesized protein is most likely to require two different signal peptides for:
 - (a) Plasma membrane
 - (b) Lysosome
 - (c) Cytosol
 - (d) Chloroplast
5. Charles Darwin's proposed conditions for natural selection encompass all of the following with the exception of:
 - (a) inheritance of both "fit" and "unfit" genes
 - (b) differential survival and reproductive success
 - (c) competition for limited resources
 - (d) overproduction of offspring
6. The first point of entry of water at the roots of a monocot plant is through the cytoplasm of cells of:
 - (a) root cap

- (b) Casparian strip
 - (c) pericycle
 - (d) endoderm
7. All of the following might be found in connective tissue EXCEPT
- (a) thrombin
 - (b) glycosaminoglycans
 - (c) collagens
 - (d) fibroblasts
8. Which of the following sets of reactions occurs in the stroma of the chloroplast in plant cells?
- (a) Calvin cycle
 - (b) Krebs cycle
 - (c) Fermentation
 - (d) Decarboxylation
9. Which of the following most accurately describes a retrotransposon?
- (a) A DNA sequence that can move from one site in the genome to another without replicating
 - (b) A DNA sequence that can be deleted from the genome without consequence
 - (c) A DNA sequence that replicates via an RNA intermediate
 - (d) A DNA sequence that replicates via a protein intermediate
10. Based on the characteristic population curves that result from plotting population growth of a species
- (a) maintain the population at a point corresponding to the midpoint of its logistic curve
 - (b) opt for zero population control once the K value of the curve has been reached
 - (c) reduce the carrying capacity of the environment to lower the K value
 - (d) increase the mortality rate
11. All of the following statements about plant embryogenesis are correct EXCEPT:
- (a) The suspensor is derived from the basal cell.
 - (b) Cotyledons are derived from the apical cell.
 - (c) Shoot apical meristem formation occurs after seed formation.
 - (d) Precursors of all three plant tissue systems are formed during embryogenesis.
12. The sight organs of crustaceans and insects contain ommatidia, which make up the individual visual

- (a) eyespot
 - (b) simple eye
 - (c) compound eye
 - (d) binocular eye
13. Annelids and arthropods are similar to each other in that members of both phyla
- (a) have segmented bodies
 - (b) have a closed circulatory system
 - (c) conduct gas exchange by diffusion through a moist membrane
 - (d) have well-developed sense organs

True / False (4 marks)

Answer True or False and justify briefly.

1. True or False: The correct answer to 'Keystone species are thought to have profound effects on t
2. True or False: The correct answer to 'Which of the following depicts the correct sequence of men

End of Paper